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WHAT IS OOGENESIS:

Oogenesis is the formation of ovum or egg.
In human oogenesis starts in the ovaries of the fetus before birth.

What are the methods of the Asexual reproduction:

the organisms reproduce by mitotic divisions producing offspring that are identical to their parents. There are some methods of Asexual reproduction: 1=Fission=> a. binary fission b. multiple fission 2=Budding (both internal and external) 3= Fragmentation 4=Parthenogenesis.

What is the pronephros:

The Pronephros appears only briefly in many vertebrate embryos, and not at all in mammalian embryos. In some vertebrates, the pronephros is the first osmoregulatory and excretory organ of the embryo as in tadpoles and other amphibian larvae.

what is the structure of the ovary:

The ovaries are small, oval-shaped glands that are located on either side of the uterus. The ovaries produce eggs and hormones. Ovary is divided into an outer cortex and an inner medulla. The cortex consists of a very cellular connective tissue stroma in which the ovarian follicles are embedded. The medulla is composed of loose connective tissue, which contains blood vessels and nerves.

Parthenogenesis and its types:

Parthenogenesis is a form of asexual reproduction in which development of egg occur without being fertilized.

Types of Parthenogenesis:

1=Diploid Parthenogenesis: In this type, no meiosis occurs and the eggs are formed by mitosis. 2=Haploid Parthenogenesis: In this type of Parthenogenesis haploid ova are formed by meiosis and they may or may not be fertilized by sperm.

Detailed note on structure of testes:

The paired testes are the male reproductive organs that produce sperms. Testes hang outside the body, the temperature inside the scrotum is about 34° C compared to a 37° C core temperature. Each testis contains over eight hundred tightly coiled seminiferous tubules which produce thousands of sperm each second in healthy young men. The walls of the seminiferous tubules are lined with two types of cells: Spermatogenic cells, which give rise to sperm, and Sustentacular cells, which nourish the sperm as they form and secrete a fluid and a hormone, inhibin, into the tubules to provide a liquid medium for the sperm. • Between the seminiferous tubules are clusters of endocrine cells, called Leydig cells, that secrete the male sex hormone, the testosterone.

Stages of development of embryo before implantation:

Four to six days after conception, the fertilized egg **cell develops** into a blastocyst and **is firmly buried** in the lining **of the uterus. At this** early stage, the embryo develops a yolk sac, which provides its first nutrients. grows, it will turn to its host (you) for **food. The tiny** villi or **tissue fingers** on the outside of the fertilized egg **cell multiply** in these **first few** weeks. **Nutrients** and remove waste. Over **time,** this **cell mass develops into** the placenta **that nourishes** and **protects** your baby throughout pregnancy.

What is the osmoregulation:

The osmoregulation is the process of regulation of water and solutes between the organisms and their environment. Or the process of regulation of internal osmotic state to external environment is called osmoregulation.

Functions:

it maintains the fluid balance and the concentration of electrolytes (salts in solution which in this case is represented by body fluid)

germs in the body:

Germs are found **in all sorts of places all** over the world. **The** four **main** types of germs are bacteria, viruses, fungi and protozoa. **They** can invade plants, animals and people and sometimes make us sick.

What is the mesonephric kidney and its function:

It is one of three excretory organs that develop in vertebrates. It is a paired organ consisting of a set of nephrons having capsules that filter blood from the glomerulus and tubules whose cells reabsorb water and nutrients and secrete nitrogenous wastes.

Function:

It serves as the main excretory organ of aquatic vertebrates and as a temporary kidney in reptiles, birds, and mammals.

What is the placenta:

The placenta is the organ that sustains the embryo and fetus throughout the pregnancy and through which gases are exchanged, nutrients are supplied and wastes are removed from the fetal systems to the maternal end.

What is the fertilization:

Fertilization is the fusion of haploid gametes, egg and sperm, to form the diploid zygote. There are two types of fertilization 1= internal fertilization and 2= external fertilization.

Structure of Human Sperm:

A mature human sperm consists of a head, midpiece, and tail.

• The head is flat, disc shaped 5.1 µm by 3.1 µm and contains the haploid nucleus, which is mostly DNA. • The acrosome, a cap over most of the head, contains an enzyme called acrosin that assists the sperm in penetrating the outer layer surrounding a secondary oocyte.

• The spiral mitochondria in the midpiece supply the ATP necessary for these movements.

• The sperm tail is 50 µm long contains an array of microtubules that bend to produce whip like movements.

vertebrate kidneys:

• Pronephros, Mesonephros, and • Metanephros.

Process of Fertilization:

Fertilization involves four events:

• (i) the sperm contacts the egg and recognition occurs.
• (ii) the sperm or sperm nucleus enters the egg.
• (iii) the egg becomes activated, and certain developmental changes begin.
• (iv) the sperm and egg nuclei fuse.